Huawei Access Node BAS-ACC-37 Basavanagudi POP BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding SOP (Alarm Code ALARM_9085, Severity Minor)

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9085

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Routing Protocol Flap on the Huawei Access Node BAS-ACC-37 at the Basavanagudi POP site, specifically encompassing BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This phenomenon is typically observed when the routing protocols fail to converge due to various reasons such as misconfiguration, hardware issues, or software bugs. In the context of the Huawei Access Node, the BGP route churn is characterized by the repeated exchange of routing information between neighboring routers, resulting in oscillations in the routing table. Similarly, OSPF adjacency loss occurs when there is a disruption in the neighbor relationship between routers, leading to a loss of route information. ISIS LSP flooding, on the other hand, is triggered by the excessive generation of Link-State Packets (LSPs), which can cause network congestion and instability. The Huawei Access Node, being a critical component of the network infrastructure, plays a pivotal role in maintaining the integrity and reliability of the network. Therefore, it is essential to promptly identify and rectify any issues that may arise in the routing protocols to prevent network disruptions and ensure seamless communication.

Pre-requisite Checks

- Verify that the Basavanagudi POP site is accessible and the network cables are securely connected.
- Confirm that the redundant link status is available and functioning correctly.
- Ensure that the Huawei Access Node BAS-ACC-37 is powered on and the fans are operational.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor relationships	High
Faulty hardware components (e.g., routers, switches)	Medium
Software bugs or compatibility issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: BGP routing table status: * Local AS: 12345 * Neighbor AS: 67890 * Routing table: 1000 entries, Description: This command is used to verify the BGP routing table status, including the local AS, neighbor AS, and the number of
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routing table entries.

Command: display ospf neighbor, Expected Output: OSPF neighbor status: * Neighbor ID: 192.168.1.1 * State: Full * Timer: 10s, Description: This command is used to verify the OSPF neighbor status, including the neighbor ID, state, and timer values.

Command: display isis lsp, Expected Output: ISIS LSP status: * LSP count: 100 * LSP interval: 10s, Description: This command is used to verify the ISIS LSP status, including the LSP count and interval values.
```

Resolution Steps

Step 1: Step: Verify BGP neighbor relationships, Command: display bgp neighbor, Description: Verify the BGP neighbor relationships and ensure that they are correctly configured.

Step 2: Step: Reset BGP routing table, Command: reset bgp routing-table, Description: Reset the BGP routing table to resolve any issues related to misconfigured BGP neighbor relationships.

Step 3: Step: Replace faulty hardware components, Command: replace router/switch, Description: Replace faulty hardware components, such as routers or switches, to resolve issues related to faulty hardware.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: BGP routing table status: * Local AS: 12345 * Neighbor AS: 67890 * Routing table: 1000 entries, Description: Verify that the BGP routing table status is stable and consistent.

Command: display ospf neighbor, Expected Output: OSPF neighbor status: * Neighbor ID: 192.168.1.1 * State: Full * Timer: 10s, Description: Verify that the OSPF neighbor status is stable and consistent.

Command: display isis lsp, Expected Output: ISIS LSP status: * LSP count: 100 * LSP interval: 10s, Description: Verify that the ISIS LSP status is stable and consistent.
```

Roll-back Procedures

- Step: Restore BGP routing table, Command: restore bgp routing-table, Description: Restore the BGP routing table to its previous state in case the fix fails.

- Step: Revert hardware changes, Command: revert router/switch, Description: Revert any hardware changes made during the fix process in case the fix fails.